

Blockchain-Based Voting System

Project Report

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Student Work Archive System
Sample Document for Testing Purposes

Abstract

This project implements a secure electronic voting system leveraging blockchain technology to ensure transparency, immutability, and verifiability of election results. The system addresses key challenges in electronic voting including voter privacy, vote integrity, and result auditability.

The implementation uses Ethereum smart contracts for vote recording and tallying, with a React-based frontend providing an accessible voting interface. Zero-knowledge proofs enable vote verification without compromising voter privacy.

System features include:

- Tamper-proof vote recording on blockchain
- Anonymous yet verifiable voting
- Real-time result tallying
- Complete audit trail for transparency
- Resistance to common attack vectors

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Chapter 1

Introduction

Electronic voting systems promise convenience and accessibility but face trust issues. Blockchain technology offers properties that address many concerns: decentralization prevents single points of failure, immutability ensures votes cannot be altered, and transparency allows public verification.

This section contains additional technical details and supporting information relevant to the chapter topic. The content demonstrates the depth of research and analysis conducted throughout this study.

Further considerations include implementation specifics, potential limitations, and suggestions for future improvements. These elements contribute to the comprehensive nature of this academic work.

Chapter 2

Background

This chapter covers blockchain fundamentals, smart contract programming, cryptographic voting schemes, and existing e-voting implementations. Analysis of previous blockchain voting projects identifies lessons learned and improvement opportunities.

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Chapter 3

System Requirements

Requirements derived from election standards include: voter authentication, vote privacy, vote integrity, universal verifiability, and resistance to coercion. Technical requirements include scalability for large elections and acceptable transaction costs.

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Chapter 4

Architecture Design

The system consists of four components: voter registration module, voting smart contract, tallying smart contract, and web frontend. The architecture separates voter identity from votes using commitment schemes.

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Chapter 5

Smart Contract Implementation

Two main contracts were developed: VoterRegistry for managing eligible voters and BallotBox for recording votes. Contracts are written in Solidity 0.8.x with security best practices. Gas optimization reduces voting cost to approximately \$0.50 per vote.

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Chapter 6

Security Analysis

The system was analyzed against common attack vectors: Sybil attacks (prevented by registration), double voting (prevented by contract logic), vote buying (mitigated by receipt-freeness), and denial of service (mitigated by gas limits).

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Chapter 7

Testing and Evaluation

Testing included unit tests for smart contracts, integration tests for the full system, and a pilot election with 200 participants. The pilot achieved 100% vote accuracy with average voting time of 45 seconds.

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Chapter 8

Conclusion

The project demonstrates blockchain viability for voting systems. Limitations include transaction costs and the need for user cryptocurrency wallets. Future work includes layer-2 scaling solutions and mobile voting applications.

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References

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